

Politecnico di Torino

Quantum Hardware Design and Optimization

Laboratory 2

Oracles Design

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Instructions for laboratories submission

Create a folder with the following name:

`gr<n>_lab2`

where *n* is the group number.
Inside, create a folder for each Exercise:

```
es01
es02
es03
```

containing the developed scripts and eventual files necessary for their execution.
The day before the laboratory evaluation meeting, submit the folder in the Elaborati section of the Portale della Didattica.

1 Design of simple oracles

Design by hand phase oracles for identifying the following items:

- 00
- 01
- 10
- 11

Now, implement the Grover Search, considering a single Grover rotation, with the designed oracles first in quirk and then in qiskit using the basic simulator.
Repeat the exercise considering two Grover rotations instead of one.

2 Design of a quantum adder

Implement a generic two-input adder block in qiskit (without carry-in) and verify its functionality, considering two input qubits for each input. Remember to verify all combinations, but for the submission leave the input at 01 and 11.
Remember to exploit quirk in the design phase for making the debugging of your implementation easier.

2.1 Optional

Now, repeat the exercise exploiting for the generic adder implementation the Takahashi advanced structure (`https://arxiv.org/abs/0910.2530`) for saving qubits (It is without the carry-in and requires a single ancilla qubit), instead of the trivial one.

3 Solve a satisfiability problem

Design a phase oracle and apply it in a proper quantum circuit to solve the following satisfiability problem.
Let's consider two inputs a and b, representable with two bits for each. Find the proper values for a and b to satisfy the following conditions:

- $a + b = 4$;
- $a \neq b$.

Exploit quirk for the design phase, then implement the solution in qiskit, choosing the proper number of Grover rotations.

3.1 Optional

Repeat the exercise by using at most six ancilla qubits.

3.2 Optional

Design a phase oracles for a three-input satisfiability problem.
Let's consider three inputs a, b, c, representable with two bits for each. Find the proper values for a, b and c to satisfy the following conditions:

- $a + b = 5$;
- $b + c = 3$;
- $a + c = 2$;
- $a \neq b \neq c$.

Exploit quirk for the design phase, then implement the solution in qiskit, choosing the proper number of Grover rotations.
Please notice that here it is fundamental to minimize the number of necessary qubits to make the circuit simulable on your PC or on the server.

Submission

It is requested to submit:

1. the offline copy of quirk circuit and the python scripts in the following format:

   ```
   es01_xx_r.py
   es01_xx_r.html
   ```

   where `xx` is the item to search and `r` the number of Grover rotations (es. `es01_00_1.py` for searching item 00 with 1 Grover rotation);

2. the python scripts for adder validation with inputs 01 and 11, named as `es02.py`;

3. the python scripts for the obtained solution, called `es03.py`.

4. for the optional exercises, the python scripts for the obtained solution, must be called, as an example, `es03_optional_1.py`.